

INDUSTRIAL SESSION

D.A.I.S.Y – Personal Offline Assistant

SUBMITTED BY:

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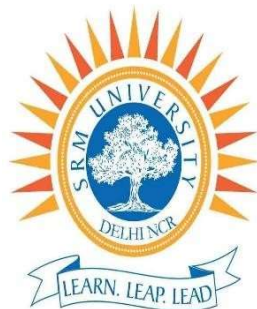
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COURSE : B.Tech CSE (DS & AI)

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CERTIFICATE

This is to certify that Killiivalavan K A has completed the Live project on the topic " D.A.I.S.Y- Personal Offline Assistant " under the guidance and supervision of Mr. Rahul Batra at SRM University.

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INTRODUCTION

DAISY is an offline voice assistant designed to address privacy concerns inherent in conventional voice assistants. By eliminating the need for internet connectivity, DAISY ensures that user data remains confidential. The system integrates advanced speech recognition and a locally hosted large language model to provide robust performance.

MOTIVATION

In an era where privacy concerns are paramount, DAISY provides a safer alternative to internet-based voice assistants. The project is motivated by the need for a secure and efficient system capable of operating in offline environments while delivering comparable functionality.

TECHNOLOGY USED

1. Programming Language: Python

- **Why Python?**

Python is chosen for its simplicity, vast library ecosystem, and strong support for AI and machine learning tasks. Its flexibility makes it ideal for rapid prototyping and development of complex applications.

2. Speech Recognition

- **Library Used:** speech_recognition

This library enables DAISY to capture and transcribe user input through a microphone. It supports multiple speech-to-text engines and provides tools to adjust for ambient noise, ensuring higher accuracy in various environments.

- **Features:**

- Multi-language support.
 - Integration with third-party speech recognition APIs like Google and Sphinx.

3. Text-to-Speech (TTS)

- **Library Used:** pyttsx3

This TTS library converts the assistant's responses into human-like speech.

- **Key Features:**

- Offline functionality.
 - Customizable voice properties such as pitch, rate, and volume.
 - Supports multiple voices (male/female).

4. Language Model

- **Locally Hosted Large Language Model (LLM): LM Studio**

The LLM is hosted locally to ensure privacy and offline functionality.

- **Why a Local LLM?**

- Privacy: No user data is sent to external servers.
 - Availability: Functions without internet connectivity.

- **Key Model Features:**

- Advanced conversational capabilities.
 - Fine-tuned for generating contextually relevant and conversational responses.

5. Hardware Requirements

- **Processor:** Ryzen 7 5800H
- **GPU:** NVIDIA 3050
- **RAM:** 16GB DDR4
 - **Why These Specifications?**
 - The computational requirements of running a locally hosted LLM and processing audio inputs demand significant hardware resources. The hardware ensures smooth operation despite the latency challenges introduced by these models.

6. Framework for LLM Integration

- **API Client:** OpenAI-compatible LM Studio API
 - Facilitates seamless integration of the local LLM with DAISY's query processing system.

7. Libraries for Integration

- **os:** Used for file handling and managing paths for audio files.
- **time:** Helps in adding delays to simulate real-time interaction and processing.

OBJECTIVES

1. To design and develop a voice assistant that prioritizes user privacy by operating entirely offline. Unlike conventional voice assistants like Siri or Google Assistant, which rely on internet-based processing, DAISY processes all data locally on the user's machine. This eliminates the risk of data breaches, unauthorized third-party access, and misuse of personal information, making it suitable for privacy-conscious users.
2. To seamlessly capture user input through voice, transcribe it into text, and process it using a locally hosted large language model (LLM). This ensures that DAISY can understand and process user queries effectively, even without cloud support. By integrating speech recognition and natural language processing, DAISY can interact conversationally, mimicking the capabilities of popular online assistants.
3. To convert DAISY's textual responses into natural-sounding speech, making the interaction intuitive and accessible. This feature ensures that DAISY can be used hands-free, providing responses in a manner that feels natural and engaging for the user.
4. To create a flexible architecture that allows for easy addition of advanced features such as contextual awareness, task management, and wake word detection. A scalable framework ensures that DAISY can evolve over time to incorporate new capabilities, making it adaptable to the changing needs of users.
5. To ensure that DAISY operates efficiently within the constraints of mid-range hardware. By focusing on performance optimization, DAISY aims to make offline voice assistant technology accessible to a wider audience, even on devices that are not equipped with high-end computational resources.

PROPOSED METHODOLOGY

The methodology for DAISY focuses on creating an offline, privacy-focused voice assistant using multiple components that work together seamlessly.

1. Data Collection and Preprocessing

- The system captures audio input from the user via the microphone and transcribes it using the speech_recognition module. The audio is then processed to remove any noise or inconsistencies before being passed to the language model.

2. Speech Recognition

- The speech_recognition module transcribes the audio input into text. This transcription process is done using offline methods to ensure data privacy. If the transcription fails, DAISY prompts the user to repeat the command.

3. Query Processing with Locally Hosted Language Model

- Once the text is transcribed, DAISY sends it to a locally hosted large language model (LLM) for processing. The model generates a relevant response based on the user's query. This avoids the need for cloud-based processing and enhances privacy.

4. Text-to-Speech Conversion

- The generated text response is converted to speech using the pyttsx3 module. This allows DAISY to speak back to the user, providing a more natural interaction.

5. Integration of Modules

- All components (speech recognition, query processing, and text-to-speech) are integrated into a seamless workflow. DAISY listens for input, processes the query, and responds without requiring internet access.

6. Performance Optimization

- The system is optimized to work efficiently on mid-range hardware by minimizing latency and reducing computational load, ensuring a smooth user experience even on limited resources.

IMPLEMENTATION

```
Listening...
Audio captured successfully.
audio written to recording.wav successfully
User: hello hello
Thinking...
DAISY: Hello! It's nice to meet you! How can I assist you today? Do you have a question or topic you'd like to chat about?
Listening...
Audio captured successfully.
audio written to recording.wav successfully
User: I want to know about Thomas Edison what do you know about him
Thinking...
DAISY: Thomas Edison is a legendary figure! He was an American inventor, businessman, and engineer who developed many groundbreaking technologies. Some of his most notable achievements include:

* Inventing the phonograph (1877), which recorded and played back sound
* Creating the first practical incandescent light bulb (1879)
* Developing the motion picture camera (1888) and improving the kinetoscope (1895)
* Improving the telegraph system, making it faster and more reliable

He's often quoted as saying "Genius is 1% inspiration and 99% perspiration." Edison held over 1,093 patents in his name and was one of the most prolific inventors in history!

Would you like to know more about a specific aspect of his life or work?
```

CONCLUSION

DAISY successfully demonstrates the potential of an offline, privacy-focused voice assistant. By processing all data locally, it eliminates the privacy concerns associated with cloud-based systems. The integration of speech recognition, natural language processing, and text-to-speech technologies ensures a seamless and intuitive user experience, even on mid-range hardware. The project provides a solid foundation for future improvements, allowing the addition of new features like contextual awareness and task management. Overall, DAISY offers a secure and efficient solution for users seeking privacy in their daily interactions with voice assistants.